

For your SarcomaAI precision oncology project, the following list outlines the key genetic and epigenetic markers identified in current sarcoma research. Sarcomas are broadly categorized into two genomic classes: **simple karyotype** (often driven by a single translocation or mutation) and **complex karyotype** (characterized by massive genomic instability and copy number alterations).

1. General Sarcoma Markers (Pan-Sarcoma)

These markers are frequently found across multiple histological types and are often used as baseline prognostic indicators.

Marker Type	Marker Detail	Clinical/Research Significance
Somatic Mutations	\$TP53\$, \$ATRX\$, \$RB1\$	The three most recurrently mutated genes across diverse sarcoma types.
Transcriptomic Signature	CINSARC (67 genes)	A complexity index signature (e.g., \$AURKA\$) that identifies high-risk tumors and predicts metastatic relapse better than histological grade.
Protein Signature	SPM6 (6 proteins)	DNA replication protein module used for risk stratification when added to nomograms.
Immune Signature	ICR (20 genes)	Immunologic Constant of Rejection signature (e.g., \$CD8A/B\$, \$CTLA4\$, \$PDCD1\$) used to predict cytotoxic immune response.
Biological Process	Necroptosis-related (10	Hub genes including

	genes)	\$RIPK1\$, \$CASP8\$, and \$RIPK3\$ that correlate with immune infiltration and prognosis.
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2. Subtype-Specific Genetic Markers

These markers are diagnostic hallmarks or prognostic drivers specific to individual sarcoma histologies.

A. Adipocytic Tumors (Liposarcomas)

- **Dedifferentiated Liposarcoma (DDLPS):**
 - **CNV Amplification:** 12q13-q15 region, specifically the \$MDM2\$ and \$CDK4\$ genes.
 - **Gene Amplification:** \$JUN\$, which is associated with worse survival.
- **Myxoid/Round Cell Liposarcoma (MLPS/MRC):**
 - **Chromosomal Translocation:** \$t(12;16)(q13;p11)\$ resulting in \$FUS::DDIT3\$ fusion, or \$t(12;22)(q13;q12)\$ for \$EWSR1::DDIT3\$.
 - **Somatic Mutation:** \$PIK3CA\$ (found in ~18% of cases), associated with \$AKT\$ activation and poor outcomes.

B. Mesenchymal & Spindle Cell Tumors

- **Synovial Sarcoma (SS):**
 - **Chromosomal Translocation:** \$t(X;18)(p11;q11)\$ resulting in \$SS18::SSX1\$, \$SS18::SSX2\$, or \$SS18::SSX4\$ fusions.
 - **Protein Marker:** \$TLE1\$ expression is a common (though not entirely specific) IHC surrogate.
- **Leiomyosarcoma (LMS):**
 - **Genomic Loss:** Deletion at chromosome 17p (affecting \$TP53\$).
 - **Somatic Mutation:** High frequency of \$RB1\$ and \$PTEN\$ mutations.
- **Undifferentiated Pleomorphic Sarcoma (UPS) & Myxofibrosarcoma (MFS):**
 - **Pathway Drivers:** Alterations in the **Hippo pathway**.
 - **Somatic Mutation:** \$NF1\$ mutations (~10.5% in MFS).
 - **Fusion (MFS):** \$TRIO::TERT\$ is often identified as a secondary oncogenic event.

C. Pediatric and Small Round Cell Tumors

- **Ewing Sarcoma:**
 - **Chromosomal Translocation:** \$t(11;22)(q24;q12)\$ leading to \$EWSR1::FLI1\$ (most common) or \$t(21;22)\$ leading to \$EWSR1::ERG\$.
- **Alveolar Rhabdomyosarcoma (ARMS):**

- **Chromosomal Translocation:** $t(2;13)(q35;q14)$ for $PAX3::FOXO1$ or $t(1;13)(p36;q14)$ for $PAX7::FOXO1$.
- **Desmoplastic Small Round Cell Tumor (DSRCT):**
 - **Chromosomal Translocation:** $t(11;22)(p13;q12)$ resulting in $EWSR1::WT1$ fusion.

D. Specialized Subtypes

- **Gastrointestinal Stromal Tumor (GIST):**
 - **Somatic Mutations:** Activating mutations in KIT (most common) or $PDGFRA$.
- **Chondrosarcoma:**
 - **Somatic Mutations:** $IDH1$ and $IDH2$ mutations, which can predict clinical outcomes and eligibility for targeted inhibitors.
- **Malignant Peripheral Nerve Sheath Tumor (MPNST):**
 - **Genomic Deletion:** $CDKN2A$ and loss of **H3K27me3** (epigenetic).

3. Epigenetic Markers

Epigenetic modifications, particularly DNA methylation, provide a "lineage-level" readout that is often more stable than mRNA expression for classification.

- **Global Methylation Profiling:** Machine learning classifiers (e.g., using Illumina 450K/EPIC arrays) can distinguish between 62 different tumor methylation classes with high accuracy (~99.9% AUC).
- **Subtype-Specific DNA Methylation:**
 - **Leiomyosarcoma Subtyping:** Distinct methylation patterns differentiate between Uterine LMS (ULMS) and Soft Tissue LMS (STLMS).
 - **Liposarcoma vs. LMS:** Significant differences identified at 7,445 CpG sites, allowing for clear separation of these subtypes.
 - **Ewing Sarcoma:** $cg00982952$ (annotated to the $GLG1$ gene) is a key methylation marker.
 - **Myxoid Liposarcoma:** $NNAT$ has been proposed as a potential tumor suppressor gene regulated by methylation in this subtype.
- **Grade-Related Methylation:** High-grade (Grade 2/3) sarcomas typically exhibit hypermethylation patterns at specific CpG sites (e.g., 144 sites identified) compared to low-grade tumors.

4. Protein (IHC) Markers as Genomic Surrogates

These are often used in clinical workflows to identify underlying genetic alterations without sequencing.

Protein Marker	Genomic Correlate	Sarcoma Subtype

MDM2 / CDK4	12q15 Amplification	WDLPS / DDLPS
SS18-SSX	\$t(X;18)\$ Fusion	Synovial Sarcoma
STAT6 (Upregulation)	\$NAB2::STAT6\$ Fusion	Solitary Fibrous Tumor
INI1 (Loss)	\$SMARCB1\$ Deletion	Epithelioid Sarcoma / poorly diff. Chordoma
SDHB (Loss)	\$SDH\$ deficiency	SDH-deficient GIST
ALK (Upregulation)	\$ALK\$ rearrangement	Inflammatory Myofibroblastic Tumor
H3K27me3 (Loss)	Epigenetic silencing	MPNST
MyoD1 / Myogenin	Myogenic program	Rhabdomyosarcoma